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Post-Training Backdoor Purification for Malware Classifiers using Dynamic Thresholding

ABSTRACT

Machine learning models are widely used for malware classification, but they are vulnerable to backdoor attacks. In a backdoor attack, attackers add hidden trigger patterns into the training data so that the model behaves normally on clean inputs but gives wrong predictions whenever the trigger appears. This project implements a post-training backdoor defense for malware classification models. The defense works without retraining the complete model from scratch and only requires a small clean dataset. The system identifies suspicious neurons responsible for the backdoor behavior, removes them, and fine-tunes the model to recover clean accuracy. The main improvement added in this project is dynamic thresholding using Median Absolute Deviation (MAD). Instead of using one fixed threshold for all layers, the threshold is calculated separately for each layer according to neuron score distribution. This makes the defense more stable and reduces unnecessary pruning. Experimental results show that the Attack Success Rate (ASR) reduces close to 0 percent while maintaining good clean accuracy.

INTRODUCTION

Machine learning models are commonly used for malware detection because they can learn patterns automatically from large datasets. However, these models are vulnerable to security attacks such as backdoor attacks. In a backdoor attack, attackers poison some training samples by adding hidden trigger patterns and changing their labels. Because of this, the model learns malicious behavior during training. The dangerous part is that the model behaves normally on clean data, making the attack difficult to detect. But whenever the trigger appears, the model gives wrong predictions according to the attacker's target. To solve this problem, this project implements a post-training purification method. The system identifies suspicious neurons responsible for backdoor behavior, removes them, and fine-tunes the model to recover performance.

PROBLEM STATEMENT

Backdoor attacks can silently compromise malware classification models during training. The infected model works normally on regular inputs but fails whenever a hidden trigger is present. Most existing defenses require the original training dataset or complete retraining, which is costly and inefficient. The goal of this project is to develop a post-training defense mechanism that can identify and remove backdoor behavior while maintaining good clean accuracy.

METHODOLOGY

The project works in three stages.

1. Dataset Poisoning

A trigger pattern is inserted into a portion of the malware dataset, and their labels are modified. The model is then trained on this poisoned dataset so that it learns both normal behavior and hidden backdoor behavior.

2. Detection and Pruning

The trained model is analyzed using a small clean dataset. Neuron activations are collected, and suspicious neurons are detected using activation scores. These neurons are then pruned from the network.

3. Fine-Tuning

After pruning, the model is fine-tuned on clean data using a small learning rate. This helps recover clean accuracy while keeping the backdoor removed.

NOVELTY ADDED

Earlier, the project used a fixed threshold after Min-Max normalization to detect suspicious neurons. The problem was that normal neurons could also cross the threshold and get removed unnecessarily. Also, smaller malicious neurons could sometimes remain undetected. To solve this, dynamic thresholding using Median Absolute Deviation (MAD) was introduced. The MAD is calculated as:

$$\text{MAD} = \text{median}(|x_i - \text{median}(x)|)$$

The threshold becomes:

$$\text{Threshold} = \text{median} + k * \text{MAD}$$

where k is usually 3. This approach automatically adjusts the threshold for each layer, reduces false positives, preserves clean accuracy better, and works well for different architectures.

IMPLEMENTATION DETAILS

The project is implemented using Python and PyTorch.

Libraries used:

- PyTorch
- NumPy
- Scikit-learn
- Matplotlib

The malware classifier is based on a Multi-Layer Perceptron (MLP) model. The project includes scripts for poisoning the dataset, training the model, purification, and final evaluation.

RESULTS AND EVALUATION

The proposed defense successfully reduced the Attack Success Rate (ASR) close to 0 percent after purification. At the same time, the model maintained good clean accuracy and overall classification performance.

Evaluation metrics used:

- Clean Accuracy (CA)
- Attack Success Rate (ASR)
- Precision
- Recall
- F1-Score

The results show that the purification process successfully removes the backdoor behavior while preserving normal malware detection performance.

LIMITATIONS

- The method requires a small clean dataset.
- Excessive pruning can reduce model accuracy.
- Performance may vary for different architectures and datasets.

CONCLUSION

This project presents a post-training backdoor defense method for malware classification models. The system identifies suspicious neurons, removes them, and fine-tunes the model to recover performance. The added MAD-based dynamic thresholding improves stability and reduces unnecessary pruning. Overall, the defense successfully reduces backdoor behavior while maintaining good clean accuracy.

REFERENCES

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